

4(a)(i)	no, nucleic acid / genetic material / DNA / RNA ; no, capsid / protein coat ; A no capsomeres only some viruses have envelopes ;	2
4(a)(ii)	heads / phosphate, (groups) of phospholipids are hydrophilic or fatty acid tails of phospholipids are hydrophobic ; A heads are hydrophilic and tails are hydrophobic heads / phosphates, point towards the centre (of the virosome so, must be aqueous / water present) ; ora for tails	2
4(b)	<i>any two from:</i> acts as a <u>non-self / foreign, antigen</u> ; triggers / stimulates, primary immune response or provides (artificial) active immunity ; (leads to) formation of antigen presenting cell ; A endocytosis / phagocytosis, to present antigen (by, macrophage neutrophil) activates, B lymphocytes / T lymphocytes ; A clonal selection formation of memory cells ;	2
4(c)	<i>any two from:</i> no change in, <u>shape</u> / tertiary structure / conformation, of active site ; ora (same) substrate able to bind to active site ; ora consequence ; e.g. so viruses can leave host cell to infect other cells the mutation is not occurring in the, gene / section of gene coding for amino acids in active site for neuraminidase ; the gene for neuraminidase is essential for survival ; AVP ;	2